Sharpe ratios, portfolios and leverage

Omitted: all third-party screenshots and publisher graphics, the original wealth-distribution chapter, bank marketing illustration, literature-result comparison and screenshot audit. Preserved: Sharpe definition, diversification, equal-risk comparisons, ideal leverage, financing-cost adjustment, hypothetical financing table, FX debt and covariance identities, measurement and risk caveats. This is not the original 36-page report.

This is a newly prepared, text-only public adaptation, not the full original research report. It retains selected explanations and mathematical derivations. Original section numbers and source-reference numbers are retained. Source links appear at the end. Interactive controls and graphics are not reproduced.

17.2 What the Sharpe ratio actually measures

The Sharpe ratio measures average excess return per unit of variability in that excess return. It can describe historical experience or a forecast; the two must be labelled separately. [27]

$x_t = R_{portfolio,t} - r_{f,t}$

$S = \text{mean}(x_t) / \text{standard deviation}(x_t)$

The benchmark, currency, return frequency and sample period must match. The benchmark is not automatically the investor’s loan rate.

Read the ratio as efficiency, not as an interest rate. A Sharpe ratio of 2 is not a 2% return and is not “twice as safe”. In the model it represents two units of excess return for each unit of volatility, measured consistently. No particular Sharpe number is a universal target or borrowing trigger.

Use total returns. Coupon or distribution yield omits market-value changes. A yield-to-maturity quote divided by a price-volatility estimate is not a properly constructed historical Sharpe ratio. A price-only series and a distribution-inclusive series must not be mixed.

Working principle:

do not rank investments by return alone, but do not replace one simplistic ranking with an unquestioned Sharpe leaderboard.

17.3 Improve the combination before increasing its size

Individual Sharpe ratios do not tell us how well assets combine. Correlation is an additional input. A holding with a lower standalone Sharpe can improve a portfolio when its returns offset other exposures. [28]

$\sigma p2 = wA2\sigma A2 + wB2\sigma B2 + 2wAwB\rho\sigma A\sigma B$

Two-asset variance identity. The covariance term, not the number of line items, drives diversification.

Diversification is about the relationship between returns:

combining imperfectly correlated assets can improve risk-adjusted performance. Sharpe ratios themselves are not additive.

Correlation: +1.0 | Expected return: 8% | Volatility: 10.00% | Sharpe: 0.60

Correlation: +0.5 | Expected return: 8% | Volatility: 8.66% | Sharpe: 0.69

Correlation: 0 | Expected return: 8% | Volatility: 7.07% | Sharpe: 0.85

Correlation: -0.5 | Expected return: 8% | Volatility: 5.00% | Sharpe: 1.20

17.4 Compare at the same risk, not just the same cash outlay

The relevant question is not simply “Which line finishes highest?” Ask whether a more efficient portfolio can meet the desired return with less volatility, or deliver more expected excess return at an equal volatility budget. That comparison requires reliable forward-looking inputs and feasible scaling. [28]

At a common volatility σ*: expected excess return = S × σ*

At a common positive excess-return target m*: volatility = m* / S

Idealised mean-variance derivation: a common risk-free rate, frictionless scaling and positive expected excess returns. Actual financing is addressed next.

Hypothetical comparison: portfolio A has expected return 8%, volatility 10% and a 2% risk-free rate, giving S = 0.60. Portfolio B has expected return 10% and volatility 20% with the same benchmark, giving S = 0.40. Scaling A to twice its exposure gives 20% volatility and 14% expected return only in the frictionless model. These are invented annual model inputs, not forecasts.

Central argument:

a lower-return portfolio is not necessarily inferior. Its return per unit of risk may leave more room for sensible exposure sizing.

17.5 Why leverage can scale a higher-Sharpe portfolio

Define own equity E, borrowing D and gross assets A = E + D. Let λ = A / E be gross exposure per unit of equity. Here D is loan principal; Part I used L for that amount. A 1× exposure is unleveraged. λ is not a loan-to-value ratio.

$$R_{equity} = \lambda R_p - (\lambda - 1) r_f$$

$$\mu_{equity} - r_f = \lambda(\mu_p - r_f)$$

$$\sigma_{equity} = \lambda \sigma_p$$

$$S_{equity} = \lambda(\mu_p - r_f) / (\lambda \sigma_p) = S_p$$

Model conditions: positive constant exposure per measurement period; borrowing at the same constant risk-free rate; no incremental costs, FX mismatch, nonlinear payoffs or forced liquidation. For a single period, the equation is a balance-sheet identity before any external capital flows. Multi-period statistics require consistent rebalancing and return construction. [27]

Unchanged Sharpe does not mean unchanged risk.

In this ideal model, both expected excess return and volatility rise in proportion to λ. Leverage scales efficiency; it does not improve the underlying ratio.

17.6 The real-world hurdle: borrowing costs and fees

A loan normally has its own price. Retain the previous one-period, single-currency structure, but set the borrowing rate rb = rf + s. Let c be incremental fees divided by opening own equity, over the same period. Asset return μp already includes any asset-level fees; do not subtract them twice.

μequity = λμp − (λ − 1)(rf + s) − c

= rf + λ(μp − rf) − (λ − 1)s − c

If funding and fees are deterministic in this simplified calculation, σequity = λσp. Dividing the excess return by that volatility gives:

Sequity = Sp − [(λ − 1)s + c] / (λσp)

Author derivation. With a positive borrowing spread and positive fees, the financed Sharpe is below the underlying Sharpe in this model. Variable rates or FX require a richer risk model.

When can the higher-Sharpe choice still win?

Let A be the lower-volatility, higher-Sharpe portfolio and B an unleveraged comparison portfolio. Scale A to B’s volatility with λ = σB / σA > 1. Then:

μlevered A − μB

= σB(SA − SB) − (λ − 1)s − c

Potential advantage: σB(SA - SB) | Must exceed: Extra financing spread and fees: (λ - 1)s + c

Potential advantage: A stronger underlying return/risk trade-off | Must exceed: The implementation costs required to reach the common risk level

Decision rule, not a recommendation:

higher Sharpe creates a potential efficiency advantage. Actual funding terms determine how much survives. Equal volatility still does not mean equal tail loss, liquidity risk or margin risk.

The derivation is symbolic. The separately labelled example below uses invented teaching inputs; no client return, current borrowing quote or suitable leverage amount is asserted.

Case: A unleveraged | Gross exposure: 1× | Borrowing rate: None | Return: 8% | Volatility: 10% | Sharpe: 0.60

Case: B unleveraged | Gross exposure: 1× | Borrowing rate: None | Return: 10% | Volatility: 20% | Sharpe: 0.40

Case: A scaled, ideal | Gross exposure: 2× | Borrowing rate: 2% | Return: 14% | Volatility: 20% | Sharpe: 0.60

Case: A scaled, spread | Gross exposure: 2× | Borrowing rate: 4% | Return: 12% | Volatility: 20% | Sharpe: 0.50

17.8 Choose the portfolio. Then set the risk budget.

Separate portfolio construction from exposure sizing. First, select a mix offering an attractive expected return for its volatility. Second, select the amount of exposure. Adding cash reduces risky exposure; borrowing increases it. Choosing a more volatile asset is not the only route to seeking a higher return. [30]

The highest-Sharpe tangent-portfolio rule depends on its assumptions: a suitable mean-variance objective, a common risk-free rate, reliable expected returns and covariances, and feasible borrowing or lending. With costly or restricted borrowing, the attainable set changes. Household liabilities and non-volatility risks also matter.

Recent research qualifies the simple rule. Moshe Levy’s 2026 study finds that realistic borrowing restrictions can substantially weaken the alignment between Sharpe rankings and investor welfare; it examines geometric-mean-based alternatives. This does not make Sharpe useless. It means the best feasible choice need not be the highest standalone Sharpe. [34]

Use this workflow:

construct → compare → size → finance → stress-test. Do not treat the tangent point as permission for unlimited borrowing.

17.10 Measure the financed position, not just the bond or fund

Three things must remain distinct: the asset’s standalone Sharpe; the financed investor-equity Sharpe; and lending headroom. The first can be attractive while the second is damaged by funding or FX, and the third can fail before a long-term thesis is realised. [6, 27]

CHF debt changes the USD investment’s return distribution

For one period with unchanged principal and interest accruing to the end, let q be USD per CHF and iCHF the contractual period interest rate. The effective USD return-cost of the debt is:

$$RD = (1 + iCHF) \times (q1 / q0) - 1$$

$$RE = \lambda RA - (\lambda - 1)RD - c$$

Author balance-sheet derivation. R_A is the USD asset total return. Costs c are measured against opening equity. Interim interest payments require their actual dates and exchange rates.

Before additional random fees and capital flows, the corresponding variance identity is:

$$Var(RE) = \lambda^2 Var(RA) + (\lambda - 1)^2 Var(RD)$$

$$- 2\lambda(\lambda - 1)Cov(RA, RD)$$

A negative covariance is dangerous here: the USD assets can fall while the USD repayment cost of CHF debt rises. A Sharpe ratio calculated from the USD bond alone misses that interaction. For an SGD-based comparison, translate the complete net position and cash flows into SGD and use a matching benchmark.

A bond coupon is not a stable price

Bond total returns incorporate market values, accrued interest and distributions. Maturity and issuer obligations matter, but the ability to hold until maturity is a separate constraint when collateral is marked along the way. A high advance ratio does not certify a high Sharpe. [6, 21]

Accounting control:

use net investor equity after liabilities. If NAV is already net of the CHF loan, do not subtract that loan a second time.

17.11 Do not mistake a smooth line for a safe investment

Sharpe can miss or obscure: Rare, severe losses | Required additional check: Drawdown, downside scenarios and loss concentration

Sharpe can miss or obscure: Illiquid or smoothed prices | Required additional check: Reliable executable valuations and liquidation time

Sharpe can miss or obscure: Changing correlations | Required additional check: Joint stress across the whole portfolio

Sharpe can miss or obscure: Funding and margin constraints | Required additional check: Haircuts, loan terms and outside liquidity

Sharpe can miss or obscure: Short samples or selected backtests | Required additional check: Out-of-sample results, robustness and estimation uncertainty

Research warning: Getmansky, Lo and Makarov show how illiquidity and smoothed returns can distort measured risk and motivate smoothing-adjusted Sharpe estimates. A steady reported NAV is not sufficient evidence of a steady economic value. [33]

Product boundary: a cap/floor payoff, an insurance illustration or a sequence of declared crediting rates cannot be assigned a superior investable Sharpe without comparable economic returns, fees, surrender terms, liquidity and liabilities. This chapter does not calculate a product-specific or client-portfolio Sharpe ratio.

No universal threshold:

a Sharpe of 2 or any other single threshold should not become an automatic borrowing trigger. A high historical number is a prompt for verification, not permission to increase debt.

17.12 Calculate it consistently, then test its reliability

A valid comparison needs comparable inputs. Use aligned total-return observations, one reporting currency, a matched benchmark and clearly defined fees. Distinguish an actual investor-equity series from a fixed-weight backtest. Do not reconstruct investment returns by tracing a picture.

Historical period Sharpe = mean(xt) / sx

Under the usual independent-observation test: t = Speriod√n

If simple annualisation is valid: Sannual = Speriod√k

Therefore: t = Sannual√(n / k)

n = number of observations; k = observations per year. Testing the mean excess return is not the same as testing raw return against zero unless the benchmark is zero. An exact classical t-test additionally assumes independent, identically distributed normal observations; large-sample approximations require suitable regularity conditions.

Andrew Lo’s research matters here: Sharpe estimates have sampling uncertainty, and serial correlation can invalidate routine square-root annualisation. A smooth-looking series can therefore produce misleading confidence and rankings. [32]

Reporting standard: show period, frequency, currency, benchmark, gross/net basis, observation count and valuation source beside every ratio. Show drawdown and margin stress separately. Use “not calculable from available data” where a required input is absent.

Use arithmetic excess-return observations for the ratio.

Report compounded growth separately; do not silently substitute CAGR into the standard Sharpe numerator.

17.13 Turn the idea into a portfolio-financing decision

The objective is a better feasible portfolio, not the largest obtainable loan. The following is a proposed decision framework, not a client allocation or a prescribed leverage ratio.

Step: 1. Establish the base portfolio | Evidence required: Holdings, weights, market values, total-return history and liabilities | Decision implication: Do not use financing to make an unsuitable asset appear attractive.

Step: 2. Compare combinations | Evidence required: Return, volatility and covariance on the same basis | Decision implication: A diversifying holding can be useful even with a lower standalone Sharpe.

Step: 3. Choose a risk budget | Evidence required: Loss capacity, cash needs, horizon and downside scenarios | Decision implication: Equal volatility is a comparison device, not a complete definition of equal risk.

Step: 4. Price financing | Evidence required: Actual rate, spread, fees, FX and hedge terms | Decision implication: Test whether the efficiency advantage survives the extra costs.

Step: 5. Test path survival | Evidence required: Stressed collateral, lower advance ratios, adverse FX and renewal terms | Decision implication: Keep repayment resources that do not depend on the same assets recovering.

Step: 6. Monitor and rebalance | Evidence required: Net equity, cash servicing, debt and headroom | Decision implication: Use actual results, not unchanged historical Sharpe, to reassess exposure.

UBS’s Singapore Lombard guidance explicitly identifies diversification, unused lending capacity and additional liquid assets as risk-mitigation measures. Those controls address funding resilience; they do not convert a volatile portfolio into a guaranteed outcome. [6]

Recommended dashboard separation

Investment efficiency: total return, excess return, volatility, portfolio Sharpe and correlation assumptions. Loss capacity: drawdown, concentrated exposures and stress losses. Financing: actual interest, debt principal, loan currency and reset dates. Survival: lending value, utilisation, collateral shortfall and independent liquidity.

Go / no-go principle:

proceed only when the asset mix, after-cost economics and funding resilience work together. An unleveraged portfolio remains a complete and legitimate choice.

Sources and scope

https://www-leland.stanford.edu/~wfsharpe/art/sr/sr.htm

https://notwins.github.io/market-intelligence-hub/research/wealth-inequality-2026-11-04/sharpe.html

No current market quotation, historical execution backtest, personalised borrowing amount or guaranteed result is supplied. A public source link is not permission to redistribute its underlying database, screenshots or report.